

A stringent discovery funnel still leaves heterogeneous evidence

TCGA-KIRC discovery; GSE40435 and GSE53757 differential-expression replication

TCGA tumor vs normal differential expression	8,534
GEO directions agree; nominal support in at least one	3,323
Adjusted survival association (FDR < 0.05)	1,186
Stage/grade direction and nominal significance	1,117
Strict effect-size and consistency filters	538
Stringent TCGA candidates (FDR < 0.01)	23

Downstream assessments are branches, not further candidate attrition

External survival (unadjusted) E-MTAB: 13/22, same-direction FDR GSE29609: 0/21, same-direction nominal	Fixed-candidate bootstrap 23/23 intervals exclude zero Selection was not repeated
Held-out prediction: separate procedure Nested selection, n = 517 Delta C = 0.0047 [-0.0079, 0.0174]	Composition and cell-source assessment Direct purity: 23/23 retain direction + FDR Marker adjustment: 17/23 retain FDR Normal-tissue atlas; cell source remains uncertain

Boxes are schematic; counts are gene symbols. External denominators are platform-mapped candidates. Bootstrap intervals condition on fixed selection. Prediction uses fold-specific selection; no clinical panel is validated.